



SOFTWARE DEV

MERN STACK DEVELOPMENT REPORT

Week 5 : State Management & Event Handling

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MERN STACK INTERN

Week 5 — Day 2

State Management & Event Handling

Overview

Today was about making the page react to clicks instead of just sitting there. In React, 'state' is a piece of data that a component remembers between renders — when state changes, React automatically redraws whatever depends on it. We used this to build an FAQ accordion where clicking a question expands its answer.

Study Topics

- Introduction to the useState hook
- Handling user events (onClick, onChange) in React

Task

Build an interactive FAQ Accordion component. Map over an array of question/answer objects and use local state to track which accordion item is expanded, ensuring only one item can be open at a time.

How We Did It (step by step)

- First created `faqData.js`, a plain array of objects, each one shaped like `{ id, question, answer }`. Keeping the content as data (not hardcoded JSX for every question) means adding a 6th question later is just adding one more object to the array — nothing else needs to change.
- Used `Array.map()` inside `FAQAccordion.jsx` to loop over that array and turn each object into one accordion row on the screen. This is the standard React pattern for turning a list of data into a list of UI elements.
- Added one piece of state: `const [openId, setOpenId] = useState(null)`. This single value stores the id of whichever question is currently open. Using one shared value (instead of a separate `true/false` for every question) is exactly what makes 'only one item open at a time' work automatically.
- Every question has an `onClick` handler that calls a small function: if the clicked question's id is already the open one, it closes it (sets it back to null); otherwise it opens the clicked one (overwriting whatever was open before, which closes the previous one for free).
- Each row checks `isOpen = item.id === openId` to decide whether to show its answer text or stay collapsed.

How the Code Works

Think of `openId` like a sticky note that can only have one name written on it at a time. When you click a question, we erase the old name and write the new one — so only the question whose name matches what's on the sticky note gets to show its answer. That's the whole trick behind 'only one open at a time', no extra logic needed.

Files / Components Created

- `src/components/accordion/faqData.js` — array of `{id, question, answer}` objects
- `src/components/accordion/FAQAccordion.jsx` — renders the list and manages open/closed state

What We Learned

`useState` is how a component remembers something between re-renders (React forgets everything and starts fresh on every normal function call — state is the exception). Storing the minimum amount of state needed (just one id, not one boolean per item) is usually the simplest and least buggy approach.

Frequently Asked Questions

What is React? —

React is a JavaScript library for building user interfaces using reusable components.

What is JSX? +

What does useState do? +

What is useEffect used for? +

Why use react-router-dom? +

Frequently Asked Questions

What is React? +

What is JSX? +

What does useState do? —

useState is a hook that lets a functional component hold and update local state.

What is useEffect used for? +

Why use react-router-dom? +